

Optimizing Learning in University: Three Evidence-Based Study Strategies

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Academic Skills (PSBE1-25)

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4 October 2026

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Introduction

As a university student, you spend many hours studying. Yet, many students are surprised and disappointed by their results. A common reason is that they rely on strategies that feel effective but are not: rereading a textbook chapter, for example, creates a sense of familiarity that is easily mistaken for real knowledge. Cognitive research has identified study techniques that produce more durable learning, yet students tend to underuse them. Self-report studies show that students favour rereading and highlighting over the methods that research supports (Miyatsu et al., 2018). Some reviews have responded by suggesting how students might improve the strategies they already prefer (Miyatsu et al., 2018) a more direct (one might even say, reasonable) route, however, is to adopt the strategies with the strongest evidence. This essay therefore asks how university students can optimize their learning, and answers by discussing three evidence-based strategies: pretesting, retrieval practice, and spaced practice. Effective studying is not simply a matter of effort, but of method.

Pretesting

The first strategy is pretesting: attempting to answer questions about material before studying it, even if the answers to the questions must be guessed. Although this seems counterintuitive, research indicates that answering questions about a topic before reading about it improves later recall (Putnam et al., 2016). The benefit arises because the attempt activates any prior knowledge and prepares the reader to connect new information with what is already known. Even incorrect guesses are useful, because they make the reader notice and remember the correct information when it appears. Prequizzing is becoming easier to practice. (Putnam et al., 2016) suggests that before a studying session (like a reading session or a lecture), a student can write down a few questions about the topic and attempt to answer them, then return to those questions afterwards to check and correct the answers. However, many textbooks also include postreading quizzed. They can also be easily repurposed for pretesting. Thirdly, the rise of AI gives us new opportunities. A chatbot can create a quiz for you after reading your lecture slides or textbook chapter. It is even possible to prompt it to follow your precise specified needs such as difficulty level. The main obstacle is psychological rather than practical: guessing at material one has not yet studied feels wrong, and students generally prefer to read first. Recognising that pretesting is a low-cost way of preparing the memory, rather than a test of current knowledge, helps to overcome this hesitation.

Retrieval Practice

The second strategy is retrieval practice: the reproduction or summarization of the study material after consuming it without looking at it.

An implementation of this strategy is the RRR-method, Read, Recite, Review, where the student is supposed to first read the study material (or visit a lecture), secondly, write down a summary from memory and lastly, review their summary, that is compare it to the study material, see what they missed or misremembered. The retrieval would be the second step of this method. The important part here is that the student not merely copies parts of the material but reproduces it from memory. This retrieval process has the advantage of both reactivating what they have learned as well as showing them what they have missed or forgotten.

From the common study strategy summarization this differs in a key way. The student will not use the material for summarization but merely write down what they remember. Thus the student will see what they did not fully understand or remember from the material, either during the recite step or when he later reviews, that is compares their summary with the actual material. From the common study strategy of notetaking it also differs in the way that it doesn't happen while studying the material but afterwards and from memory.

The common study strategies discussed in (Miyatsu et al., 2018) all don't make use of retrieval.

When practicing highlighting, a student merely alters the material while studying.

Similarly, notetaking does not use any retrieval practices.

In classical summarization, the student writes down key points while studying the material. However, as practices in the RRR-method, the small modification of summarizing after consumption of the material without using it, makes this common strategy a retrieval practice.

Similarly, unlike classic outlining, if used without the material by hand, it could be used as a retrieval skill with the caveat that outlining is a skill that must be learned and practiced to be mastered.

Spaced Practice

The third strategy is spaced repetition: this has nothing to do with how to study but with when to study. If a student practices spaced repetition, they study for short periods of time, mixes studying of different subjects and takes considerable breaks between study sessions.

The opposite of spaced repetition is massed crammed studying.

According to (Putnam et al., 2016), for the same time spent studying, spaced repetition gives much better results.

The typical example of cramming is a long study session the day even night before an exam.

The difference in retrieval become even greater in the long term.

So while this might even give acceptable results on the day of the exam, for long-term learning, spaced repetition is the much better study strategy.

Assuming university students are not merely studying to pass exams but are eager to learn and remember, spaced repetition is the far superior strategy.

Conclusion

References

Bibliography

Miyatsu, T., Nguyen, K., & McDaniel, M. A. (2018). Five Popular Study Strategies: Their Pitfalls and Optimal Implementations. *Perspectives on Psychological Science*, 13(3), 390–407. <https://doi.org/10.1177/1745691617710510>

Putnam, A. L., Sungkhasettee, V. W., & Roediger, L., Henry L. (2016). Optimizing Learning in College: Tips From Cognitive Psychology. *Perspectives on Psychological Science*, 11(5), 652–660. <https://doi.org/10.1177/1745691616645770>

AI disclosure statement

An AI assistant was used extensively in the preparation of this essay, and this is disclosed in full. The assistant was opencode, running a DeepSeek language model. It drafted the initial text of the essay on the basis of the two assigned articles and the assignment brief, and it was also used to plan the structure of the essay and to discuss which study strategies to include. Every prompt sent to the assistant, and every reply it gave, are recorded in the project's public repository, together with a word-level attribution of the final text that shows which passages originated with the assistant and which were written or rewritten by the author. All sources cited in the essay were checked against the original articles, and the author reviewed and edited the final text and takes responsibility for its content.

Reflection

Writing this essay was more instructive than I expected. The two articles described study strategies that I have rarely used, and one in particular stood out: the read–recite–review method. Reading about it, I realised that I had planned to recite and review the articles after reading them, but never did, and I am now fairly sure that this is why I remember so little of their detail. The exercise also made me notice how often I choose study methods because they feel productive rather than because they work. In future, I want to try retrieval practice and spacing deliberately, and to pay attention to whether they feel harder at first. The writing itself was manageable, but I found it difficult to keep the text objective and to decide how much detail each strategy needed within the word limit.